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CRYPTOGRAPHY

MODULE 20

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1. INTRODUCTION TO CRYPTOGRAPHY

1. Introduction to Cryptography

1.1 DEFINITION OF CRYPTOGRAPHY

Cryptography is the science of **protecting information** by transforming it into an **unreadable format** so that only authorized parties can access it.

In cybersecurity, cryptography ensures:

- Confidentiality
- Integrity
- Authentication
- Non-repudiation

CEH focuses on **how cryptography works, where it fails, and how attackers exploit weak implementations.**

2. OBJECTIVES OF CRYPTOGRAPHY

The primary objectives of cryptography are:

2.1 Confidentiality

Ensures that data is **only accessible to authorized users.**

Example:

- Encrypting sensitive files
 - HTTPS communication
-

2.2 Integrity

Ensures data is **not altered** during transmission or storage.

Example:

- Hashing
 - Message Authentication Codes (MAC)
-

1. INTRODUCTION TO CRYPTOGRAPHY

2.3 Authentication

Verifies the identity of users or systems.

Example:

- Digital certificates
 - Public key authentication
-

2.4 Non-Repudiation

Prevents a sender from denying an action.

Example:

- Digital signatures
-

3. IMPORTANCE OF CRYPTOGRAPHY IN CYBERSECURITY

Cryptography is used in:

- Secure communications (TLS/SSL)
- Data-at-rest protection
- Digital identities
- Secure authentication systems

Without cryptography:

- Data can be intercepted
- Messages can be modified
- Identities can be spoofed

1. INTRODUCTION TO CRYPTOGRAPHY

4. CRYPTOGRAPHIC TERMINOLOGY (CEH EXAM FOCUS)

Term	Description
Plaintext	Original readable data
Ciphertext	Encrypted unreadable data
Encryption	Converting plaintext to ciphertext
Decryption	Converting ciphertext to plaintext
Cipher	Algorithm used for encryption
Key	Secret value controlling encryption

5. TYPES OF CRYPTOGRAPHY

5.1 Symmetric Cryptography

- Same key for encryption and decryption
- Fast and efficient

5.2 Asymmetric Cryptography

- Uses public and private key pairs
- Slower but more secure for key exchange

5.3 Hash Functions

- One-way transformation
- Used for integrity verification

(Each will be covered in **separate detailed parts.**)

6. CRYPTOGRAPHIC ATTACKS

Attackers target cryptography by:

- Breaking weak algorithms
- Exploiting poor key management

1. INTRODUCTION TO CRYPTOGRAPHY

- Abusing implementation flaws

Cryptography often fails **not because of algorithms**, but because of **human error and misconfiguration**.

7. RISKS OF WEAK CRYPTOGRAPHY

- Data breaches
 - Credential theft
 - Man-in-the-Middle attacks
 - Legal and compliance failures
-

8. MITRE ATT&CK MAPPING

Activity	MITRE Tactic	Technique ID	Description
Weak Encryption Abuse	Credential Access	T1552	Unsecured Credentials
MITM	Credential Access	T1557	Man-in-the-Middle
Certificate Abuse	Defense Evasion	T1553	Subvert Trust Controls

9. CEH Exam-Oriented Notes (Part 1)

- Cryptography ≠ security by itself
- Algorithms are strong; implementations fail
- Key management is critical
- Hashing ≠ encryption
- Public key ≠ private key

2: CLASSICAL CRYPTOGRAPHY & EARLY ENCRYPTION TECHNIQUES

2: Classical Cryptography & Early Encryption Techniques

1. INTRODUCTION TO CLASSICAL CRYPTOGRAPHY

Classical cryptography refers to **early encryption techniques** developed before modern computers. These techniques form the **foundation of modern cryptography** and are still tested in CEH to understand **encryption logic and cryptanalysis**.

Classical ciphers rely on:

- Character substitution
 - Character rearrangement
 - Simple mathematical operations
-

2. SUBSTITUTION CIPHERS

2.1 Definition

A substitution cipher replaces each character in plaintext with another character according to a fixed rule.

2.2 Monoalphabetic Substitution Cipher

Working Principle

- One-to-one mapping between plaintext and ciphertext alphabets
- Same substitution throughout the message

Example

Plaintext: ATTACK

Ciphertext: QZZQEA

Weaknesses

- Frequency analysis easily breaks it
- Patterns remain visible

2: CLASSICAL CRYPTOGRAPHY & EARLY ENCRYPTION TECHNIQUES

2.3 Caesar Cipher

Definition

A Caesar cipher shifts characters by a fixed number of positions.

Example (Shift = 3)

Plaintext: HELLO

Ciphertext: KHOOR

Advantages

- Simple
- Easy to implement

Disadvantages

- Only 25 possible keys
 - Easily brute-forced
-

2.4 Risks

- No security against modern attacks
 - Educational use only
-

3. POLYALPHABETIC SUBSTITUTION CIPHER

3.1 Vigenère Cipher

Working Principle

- Uses a keyword
- Multiple substitution alphabets

Example

Plaintext: ATTACK

Key: LEMON

2: CLASSICAL CRYPTOGRAPHY & EARLY ENCRYPTION TECHNIQUES

Ciphertext: LXFOPV

Strengths

- Reduces frequency patterns

Weaknesses

- Vulnerable to Kasiski examination
-

4. TRANSPOSITION CIPHERS

4.1 Definition

Transposition ciphers rearrange characters **without changing them**.

4.2 Rail Fence Cipher

Example

Plaintext: HELLOWORLD

Ciphertext: HLOOLELWRD

Weakness

- Letter frequencies remain unchanged
-

5. PRODUCT CIPHERS

5.1 Definition

Combination of:

- Substitution
- Transposition

Modern ciphers (AES, DES) are **product ciphers**.

2: CLASSICAL CRYPTOGRAPHY & EARLY ENCRYPTION TECHNIQUES

6. CLASSICAL CRYPTANALYSIS

6.1 Frequency Analysis

- Analyzes letter frequency
 - Most common English letter = E
-

6.2 Known Plaintext Attack

- Attacker knows part of plaintext
-

7. RISKS & LIMITATIONS OF CLASSICAL CRYPTOGRAPHY

- No key management
 - No authentication
 - Easily breakable
-

8. MITRE ATT&CK MAPPING

Concept	MITRE Tactic	Technique
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Weak Encryption Credential Access	T1552	
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Brute Force	Credential Access T1110	
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9. CEH Exam-Oriented Notes

- Caesar cipher uses fixed shift
- Vigenère uses keyword
- Transposition ≠ substitution
- Frequency analysis breaks monoalphabetic ciphers
- Classical ciphers are **not secure today**

3: MODERN SYMMETRIC-KEY CRYPTOGRAPHY

3: Modern Symmetric-Key Cryptography

1. INTRODUCTION TO SYMMETRIC-KEY CRYPTOGRAPHY

Symmetric-key cryptography uses the **same secret key** for both encryption and decryption. It is widely used due to its **speed and efficiency**, especially for encrypting large volumes of data.

In CEH, focus is placed on:

- Algorithms
 - Key sizes
 - Strengths and weaknesses
 - Real-world usage
-

2. CHARACTERISTICS OF SYMMETRIC CRYPTOGRAPHY

- Single shared secret key
 - Fast encryption and decryption
 - Requires secure key distribution
 - Less computational overhead
-

3. BLOCK CIPHERS VS STREAM CIPHERS

3.1 Block Ciphers

- Encrypt fixed-size blocks of data
- Common block size: 64-bit or 128-bit

Examples:

- DES
- AES

3: MODERN SYMMETRIC-KEY CRYPTOGRAPHY

3.2 Stream Ciphers

- Encrypt data bit-by-bit or byte-by-byte

Examples:

- RC4
-

4. DATA ENCRYPTION STANDARD (DES)

4.1 Overview

DES is a **64-bit block cipher** with a **56-bit key** developed in the 1970s.

4.2 Working Principle

- 16 rounds of encryption
 - Feistel structure
-

4.3 Weaknesses

- Short key length
 - Vulnerable to brute-force attacks
-

4.4 Current Status

- Considered insecure
 - Deprecated
-

5. TRIPLE DES (3DES)

5.1 Overview

3DES applies DES **three times** with different keys.

3: MODERN SYMMETRIC-KEY CRYPTOGRAPHY

5.2 Strengths

- Improved security over DES
-

5.3 Weaknesses

- Slow
 - Vulnerable to meet-in-the-middle attacks
-

5.4 Current Status

- Being phased out
-

6. ADVANCED ENCRYPTION STANDARD (AES)

6.1 Overview

AES is the current **industry-standard symmetric encryption algorithm**.

6.2 Key Sizes

- 128-bit
 - 192-bit
 - 256-bit
-

6.3 Strengths

- Strong security
 - Efficient
 - Resistant to known attacks
-

6.4 Usage

- Disk encryption

3: MODERN SYMMETRIC-KEY CRYPTOGRAPHY

- VPNs
 - HTTPS
-

7. MODES OF OPERATION

7.1 ECB (Electronic Codebook)

- Identical plaintext blocks → identical ciphertext
 - **Not secure**
-

7.2 CBC (Cipher Block Chaining)

- Uses IV
 - More secure than ECB
-

7.3 GCM (Galois/Counter Mode)

- Provides confidentiality + integrity
 - Preferred mode
-

8. KEY MANAGEMENT CHALLENGES

- Secure key distribution
- Key storage
- Key rotation

Key compromise = data compromise.

9. SYMMETRIC CRYPTOGRAPHY ATTACKS

- Brute-force
- Known plaintext

3: MODERN SYMMETRIC-KEY CRYPTOGRAPHY

- Weak mode exploitation
-

10. MITRE ATT&CK MAPPING

Concept	MITRE Tactic	Technique ID
---------	--------------	--------------

Weak Cipher Use	Credential Access	T1552
-----------------	-------------------	-------

Key Exposure	Credential Access	T1555
--------------	-------------------	-------

11. CEH Exam-Oriented Notes (Part 3)

- Same key for encryption & decryption
- DES = insecure
- AES = standard
- ECB mode is weak
- GCM provides integrity

4: ASYMMETRIC (PUBLIC-KEY) CRYPTOGRAPHY

4: Asymmetric (Public-Key) Cryptography

1. INTRODUCTION TO ASYMMETRIC CRYPTOGRAPHY

Asymmetric cryptography uses **two mathematically related keys**:

- **Public Key** (shared openly)
- **Private Key** (kept secret)

It solves the **key distribution problem** present in symmetric cryptography and is widely used for **secure key exchange, authentication, and digital signatures**.

2. CHARACTERISTICS OF ASYMMETRIC CRYPTOGRAPHY

- Two different keys
 - Slower than symmetric encryption
 - Enables secure communication over untrusted networks
 - Used together with symmetric cryptography in real systems
-

3. PUBLIC-KEY INFRASTRUCTURE (PKI)

3.1 Definition

PKI is a framework that manages:

- Digital certificates
 - Public-private key pairs
 - Trust relationships
-

3.2 Core PKI Components

- Certificate Authority (CA)
 - Registration Authority (RA)
 - Digital certificates
-

4: ASYMMETRIC (PUBLIC-KEY) CRYPTOGRAPHY

- Certificate revocation lists (CRL)
-

4. RSA (RIVEST–SHAMIR–ADLEMAN)

4.1 Overview

RSA is the **most widely used asymmetric encryption algorithm**.

4.2 Working Principle

- Based on factorization of large prime numbers
 - Key pair generated using mathematical operations
-

4.3 Key Sizes

- 1024-bit (deprecated)
 - 2048-bit (standard)
 - 4096-bit (high security)
-

4.4 Strengths

- Well-studied
 - Widely supported
-

4.5 Weaknesses

- Slow
 - Vulnerable if key size is small
 - Poor random number generation can break security
-

4: ASYMMETRIC (PUBLIC-KEY) CRYPTOGRAPHY

5. DIFFIE-HELLMAN KEY EXCHANGE

5.1 Definition

Diffie-Hellman allows two parties to **securely exchange a symmetric key** over an insecure channel.

5.2 Characteristics

- Does not encrypt data directly
 - Used for key agreement
-

5.3 Weaknesses

- Vulnerable to MITM without authentication
-

6. ELLIPTIC CURVE CRYPTOGRAPHY (ECC)

6.1 Overview

ECC provides **strong security with smaller key sizes**.

6.2 Advantages

- Faster than RSA
 - Smaller keys
 - Lower power consumption (ideal for mobile)
-

6.3 Usage

- Mobile devices
- IoT
- TLS

4: ASYMMETRIC (PUBLIC-KEY) CRYPTOGRAPHY

7. DIGITAL SIGNATURES

7.1 Purpose

- Authentication
 - Integrity
 - Non-repudiation
-

7.2 Working

1. Hash message
 2. Encrypt hash with private key
 3. Verify using public key
-

8. HYBRID CRYPTOSYSTEMS

Real-world systems combine:

- Asymmetric cryptography (key exchange)
- Symmetric cryptography (data encryption)

Example:

- HTTPS (TLS)
-

9. ASYMMETRIC CRYPTOGRAPHY ATTACKS

- Man-in-the-Middle
- Weak key sizes
- Poor certificate validation

4: ASYMMETRIC (PUBLIC-KEY) CRYPTOGRAPHY

10. MITRE ATT&CK MAPPING

Concept	MITRE Tactic	Technique ID
Certificate Abuse	Defense Evasion	T1553
MITM	Credential Access	T1557
Weak Key Use	Credential Access	T1552

11. CEH Exam-Oriented Notes

- Public key $\neq$ private key
- RSA used for encryption & signatures
- Diffie-Hellman = key exchange
- ECC = strong security, small keys
- Asymmetric is slower than symmetric

5: HASH FUNCTIONS & MESSAGE INTEGRITY

5: Hash Functions & Message Integrity

1. INTRODUCTION TO HASH FUNCTIONS

Hash functions are cryptographic algorithms that transform input data of **any size** into a **fixed-length output**, called a **hash value** or **message digest**.

Hashing is primarily used to ensure:

- Data integrity
 - Password protection
 - Digital signatures
-

2. CHARACTERISTICS OF CRYPTOGRAPHIC HASH FUNCTIONS

A secure hash function must satisfy:

- **Deterministic** – Same input → same hash
 - **Fast computation**
 - **Preimage resistance** – Cannot derive input
 - **Collision resistance** – No two inputs produce same hash
 - **Avalanche effect** – Small input change → large output change
-

3. COMMON HASH ALGORITHMS

3.1 MD5 (Message Digest 5)

Overview

- Produces 128-bit hash
- Widely used historically

Weaknesses

- Collision attacks exist
 - Considered broken
-

5: HASH FUNCTIONS & MESSAGE INTEGRITY

Current Status

✗ Not secure for cryptographic use

3.2 SHA-1 (Secure Hash Algorithm 1)

Overview

- Produces 160-bit hash

Weaknesses

- Vulnerable to collision attacks

Current Status

✗ Deprecated

3.3 SHA-2 Family

Includes:

- SHA-256
- SHA-384
- SHA-512

Strengths

- Strong collision resistance
- Widely adopted

Usage

- Digital signatures
 - TLS
 - File integrity
-

3.4 SHA-3

- Based on Keccak algorithm

5: HASH FUNCTIONS & MESSAGE INTEGRITY

- Alternative to SHA-2
 - Resistant to length extension attacks
-

4. HASHING PASSWORDS

4.1 Password Storage

- Passwords are stored as hashes
 - Prevents plaintext password storage
-

4.2 Salting

- Random value added before hashing
 - Prevents rainbow table attacks
-

4.3 Key Stretching

- Increases computational cost
 - Examples: PBKDF2, bcrypt
-

5. MESSAGE AUTHENTICATION CODE (MAC)

5.1 Definition

MAC ensures:

- Integrity
 - Authentication
-

5.2 HMAC

- Combines hash function + secret key
- Resistant to tampering

5: HASH FUNCTIONS & MESSAGE INTEGRITY

6. HASH-BASED ATTACKS

6.1 Collision Attack

- Two different inputs → same hash
-

6.2 Birthday Attack

- Exploits probability of collisions
-

6.3 Rainbow Table Attack

- Precomputed hashes for password cracking
-

7. RISKS OF WEAK HASHING

- Password compromise
 - Forged digital signatures
 - Data integrity loss
-

8. MITRE ATT&CK MAPPING

CONCEPT	MITRE TACTIC	TECHNIQUE ID
WEAK HASH USE	Credential Access	T1552
PASSWORD CRACKING	Credential Access	T1110

9. CEH Exam-Oriented Notes

- Hashing ≠ encryption
- MD5 and SHA-1 are broken
- SHA-256 is widely used
- Salting prevents rainbow tables

6: CRYPTOGRAPHIC ATTACKS & VULNERABILITIES

6: Cryptographic Attacks & Vulnerabilities

1. INTRODUCTION TO CRYPTOGRAPHIC ATTACKS

Cryptographic attacks exploit **weak algorithms, poor implementations, improper configurations, and human errors** rather than breaking strong algorithms mathematically.

CEH emphasizes understanding **how crypto fails in real-world systems**.

2. BRUTE-FORCE ATTACK

2.1 Definition

An attacker tries **all possible keys or passwords** until the correct one is found.

2.2 Factors Affecting Brute Force

- Key length
 - Key randomness
 - Computational power
-

2.3 Mitigation

- Strong key lengths
 - Rate limiting
 - Account lockout
-

3. DICTIONARY ATTACK

3.1 Definition

Uses a predefined list of common passwords or phrases.

3.2 Mitigation

- Strong password policies
-

6: CRYPTOGRAPHIC ATTACKS & VULNERABILITIES

- Salting & hashing
-

4. MAN-IN-THE-MIDDLE (MITM) ATTACK

4.1 Definition

Attacker intercepts and possibly alters communication between two parties.

4.2 Cryptographic Context

- Exploits weak key exchange
 - Poor certificate validation
-

4.3 Mitigation

- Certificate validation
 - Mutual authentication
-

5. KNOWN PLAINTEXT ATTACK

5.1 Definition

Attacker knows part of the plaintext and ciphertext.

5.2 Mitigation

- Strong algorithms
 - Random IVs
-

6. CHOSEN PLAINTEXT ATTACK

6.1 Definition

Attacker chooses plaintexts and observes ciphertexts.

6: CRYPTOGRAPHIC ATTACKS & VULNERABILITIES

7. PADDING ORACLE ATTACK

7.1 Definition

Exploits padding errors in block cipher modes.

7.2 Mitigation

- Use authenticated encryption (GCM)
 - Generic error messages
-

8. SIDE-CHANNEL ATTACKS

8.1 Types

- Timing attacks
 - Power analysis
 - Electromagnetic leaks
-

8.2 Mitigation

- Constant-time algorithms
 - Hardware protections
-

9. WEAK RANDOM NUMBER GENERATION

9.1 Risk

Predictable keys and IVs.

9.2 Mitigation

- Cryptographically secure RNG

6: CRYPTOGRAPHIC ATTACKS & VULNERABILITIES

10. MITRE ATT&CK MAPPING

Attack	MITRE Tactic	Technique ID
Brute Force	Credential Access	T1110
MITM	Credential Access	T1557
Weak Crypto	Defense Evasion	T1552

11. CEH Exam-Oriented Notes

- Crypto fails due to implementation
- MITM targets key exchange
- Padding oracle exploits error messages
- Side-channel ≠ algorithm break
- RNG quality is critical

Cryptography Best Practices

1.1 Use Strong, Modern Algorithms

- AES-128/256 for symmetric encryption
 - RSA-2048+ or ECC for asymmetric
 - SHA-256 or higher for hashing
-

1.2 Secure Key Management

- Protect private keys
- Rotate keys periodically
- Use Hardware Security Modules (HSMs)

1.3 Use Authenticated Encryption

- Prefer AES-GCM or ChaCha20-Poly1305
 - Prevents tampering and padding oracle attacks
-

1.4 Proper Certificate Management

- Validate certificates
 - Use trusted CAs
 - Implement certificate revocation
-

2. CRYPTOGRAPHIC STANDARDS

2.1 Common Standards

- AES (FIPS 197)
 - RSA (PKCS #1)
-

CRYPTOGRAPHY BEST PRACTICES

- SHA (FIPS 180)
 - TLS (RFC 8446)
-

2.2 Compliance Frameworks

- PCI-DSS
 - HIPAA
 - GDPR
-

3. REAL-WORLD CRYPTOGRAPHY FAILURES (CASE STUDIES)

3.1 Case Study 1: Weak SSL/TLS Configuration

Issue

- Outdated TLS version
- Weak cipher suites

Impact

- MITM attacks
- Data interception

Lesson

- Disable legacy protocols
-

3.2 Case Study 2: Password Hash Breach

Issue

- MD5 used without salt

Impact

- Mass password cracking

Lesson

- Use salted, strong hashes
-

CRYPTOGRAPHY BEST PRACTICES

3.3 Case Study 3: Random Number Generator Failure

Issue

- Predictable RNG

Impact

- Private key recovery

Lesson

- Use CSPRNG
-

4. CRYPTOGRAPHY IN CEH ATTACK LIFECYCLE

Phase	Crypto Aspect
Reconnaissance	Certificate analysis
Exploitation	Weak crypto abuse
Post-Exploitation	Credential decryption
Covering Tracks	Encrypted exfiltration

5. Complete MITRE ATT&CK Mapping

Area	Tactic	Technique ID
Weak Crypto	Credential Access	T1552
MITM	Credential Access	T1557
Key Theft	Credential Access	T1555

6. CEH Exam Strategy – Cryptography Module

6.1 High-Probability Exam Topics

- AES vs DES vs 3DES

CRYPTOGRAPHY BEST PRACTICES

- RSA vs ECC
 - Hashing vs encryption
 - Salting & HMAC
 - MITM and certificate attacks
-

6.2 Common Exam Traps

- Confusing hashing with encryption
 - Assuming algorithms fail instead of implementations
 - Ignoring key management
-

7. Quick Revision

Table Concept Key

Point

AES	Symmetric standard
RSA	Public-key encryption
ECC	Small key, strong
SHA-256	Secure hashing
GCM	Authenticated encryption

8. Best Defensive Crypto Architecture

- Hybrid encryption
- Strong key management
- Regular audits
- Secure coding practices

CRYPTOGRAPHY BEST PRACTICES

CONCLUSION

Cryptography is the **backbone of modern cybersecurity**, but its strength depends on **correct implementation, key management, and configuration**. CEH emphasizes understanding cryptography not just to secure systems, but to recognize **how attackers exploit weak cryptographic designs**.

Mastering this module equips you to:

- Identify weak crypto implementations
 - Prevent MITM and credential theft
 - Design secure communication systems
-

REFERENCES

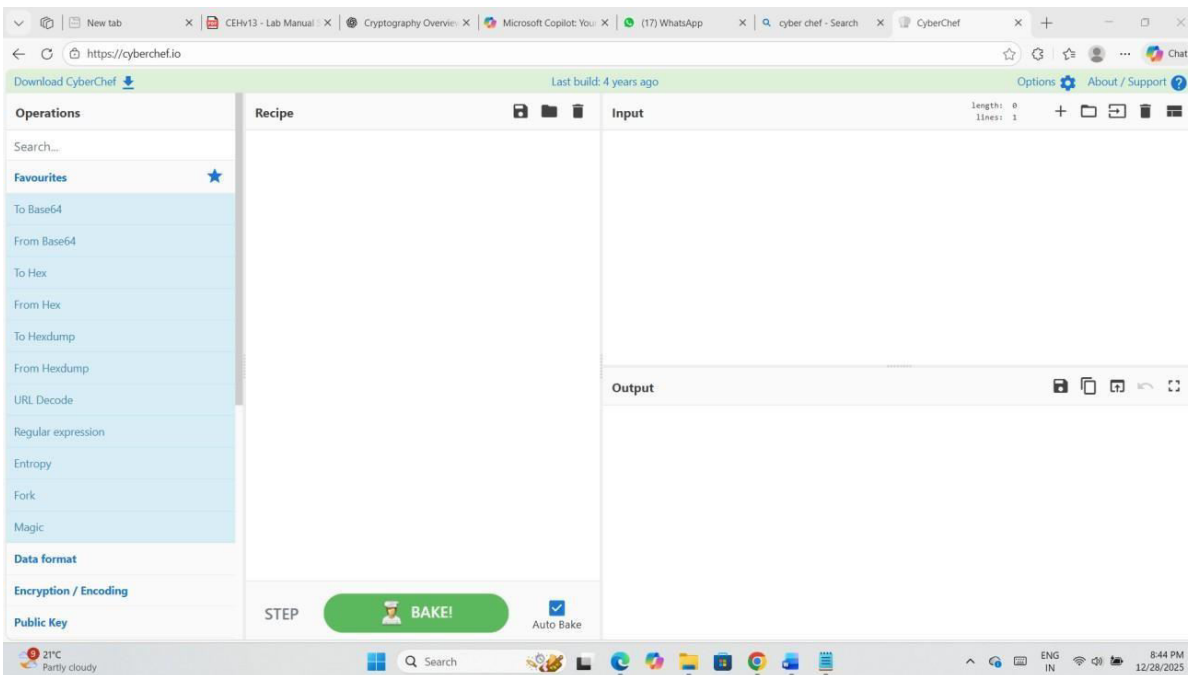
1. EC-Council CEHv13 Official Curriculum
2. NIST Cryptographic Standards
3. MITRE ATT&CK Framework
4. OWASP Cryptographic Storage Guide
5. RFCs for TLS and PKI

POC's

1. CYBERCHEF

CyberChef is a versatile browser-based tool that allows you to perform a wide range of “cyber” operations. It supports everything from simple text manipulations to advanced cryptographic functions, including various **hashing algorithms** such as MD5, SHA-1, SHA-256, SHA-512, and **encoding techniques** like text-to-hexadecimal, binary, Base64, or URL encoding.

Multi-layer hashing involves applying multiple hash functions sequentially to data. Instead of using a single hash function, the output of one hash function is used as the input for the next, creating a layered or nested structure that enhances data security. Link : <https://cyberchef.io/>



POC'S

MD5 hash

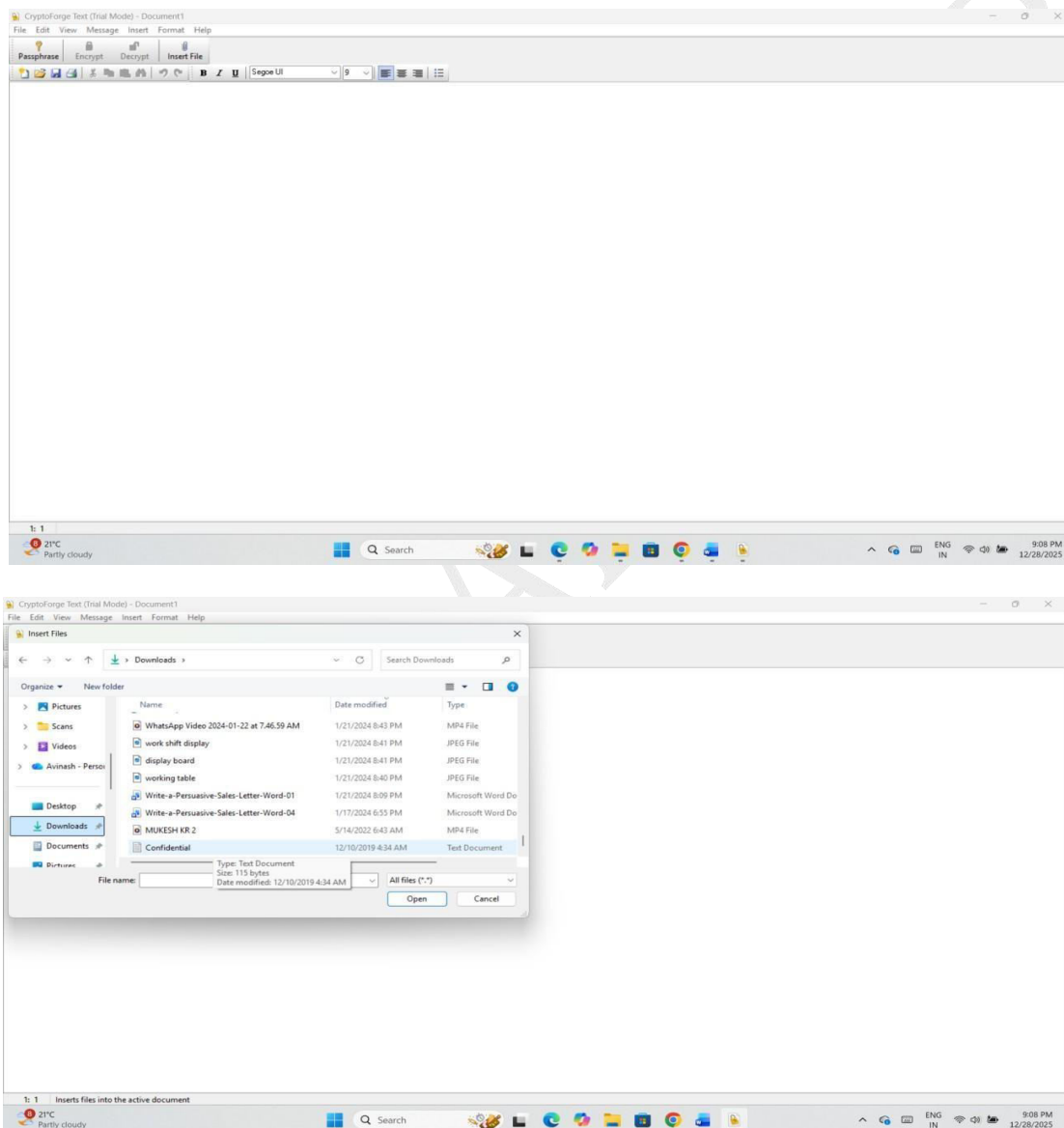
The screenshot shows the CyberChef web application interface. The browser address bar displays `https://cyberchef.io/#recipe=MD5()`. On the left sidebar, the 'Generate all hashes' section is active, with 'MD5' selected. The 'Recipe' panel shows 'MD5' as the active recipe. The 'Input' panel shows a file upload dialog for 'CTI QB (2).pdf' (114,831 bytes). The 'Output' panel displays the resulting MD5 hash: `635de617dca103d9760876a1a5b8bc74`. The interface includes a 'BAKE!' button and an 'Auto Bake' checkbox.

By using both encryption

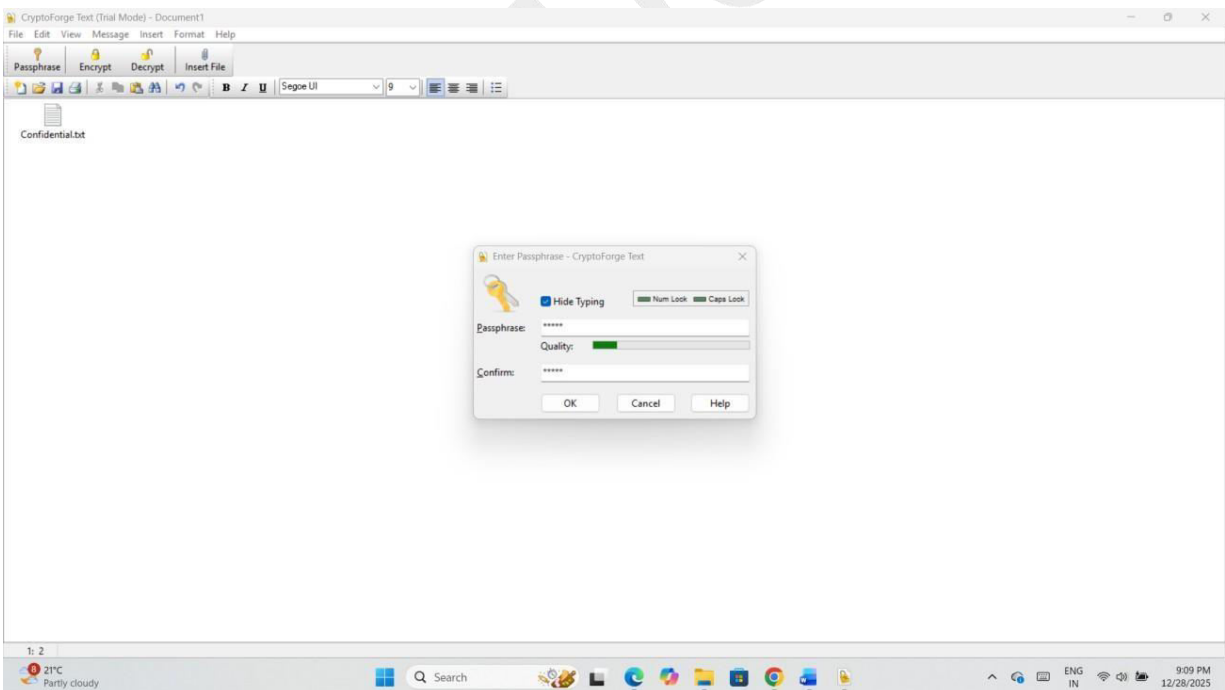
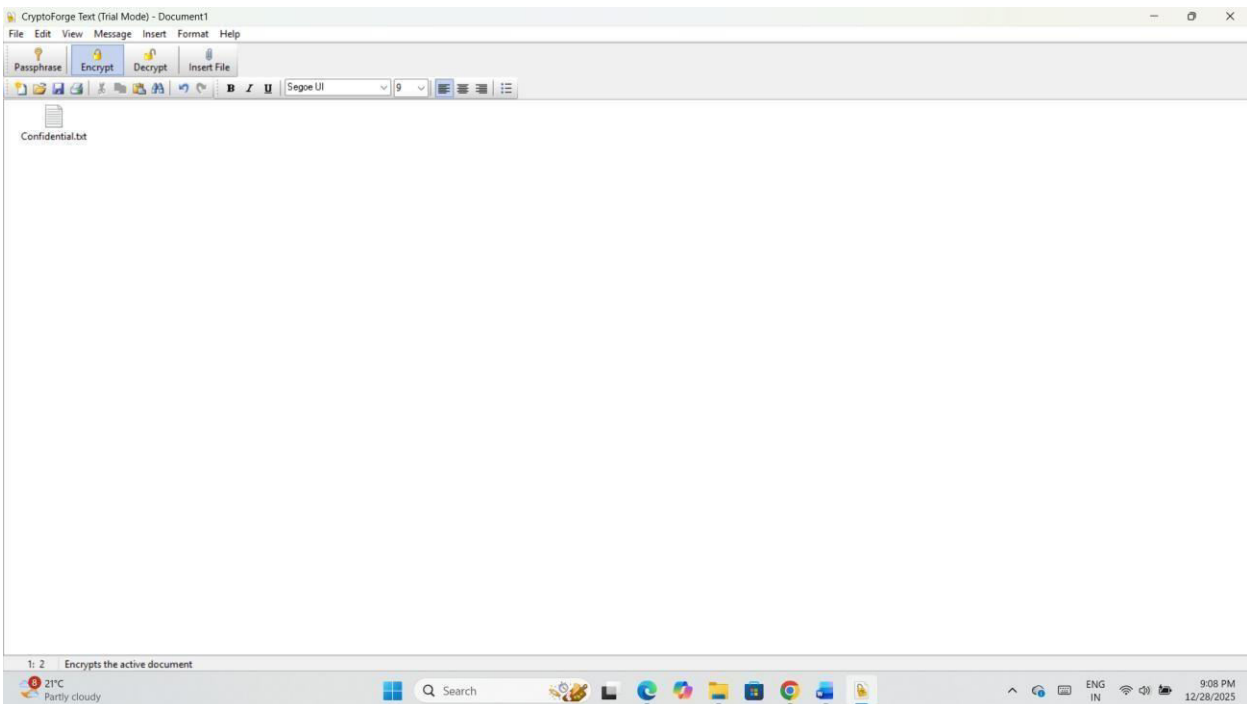
The screenshot shows the CyberChef web application interface. The browser address bar displays `https://cyberchef.io/#recipe=MD2(18)SHA1(80)`. On the left sidebar, the 'Hashing' section is active, with 'MD2' and 'SHA1' selected. The 'Recipe' panel shows 'MD2' (18 rounds) and 'SHA1' (80 rounds) as the active recipes. The 'Input' panel shows a file upload dialog for 'resume sample.pdf' (123,057 bytes). The 'Output' panel displays the resulting combined hash: `7c4d7923e16b21fbc563c627ac52650afbc8462`. The interface includes a 'BAKE!' button and an 'Auto Bake' checkbox.

2. CRYPTOFORGE

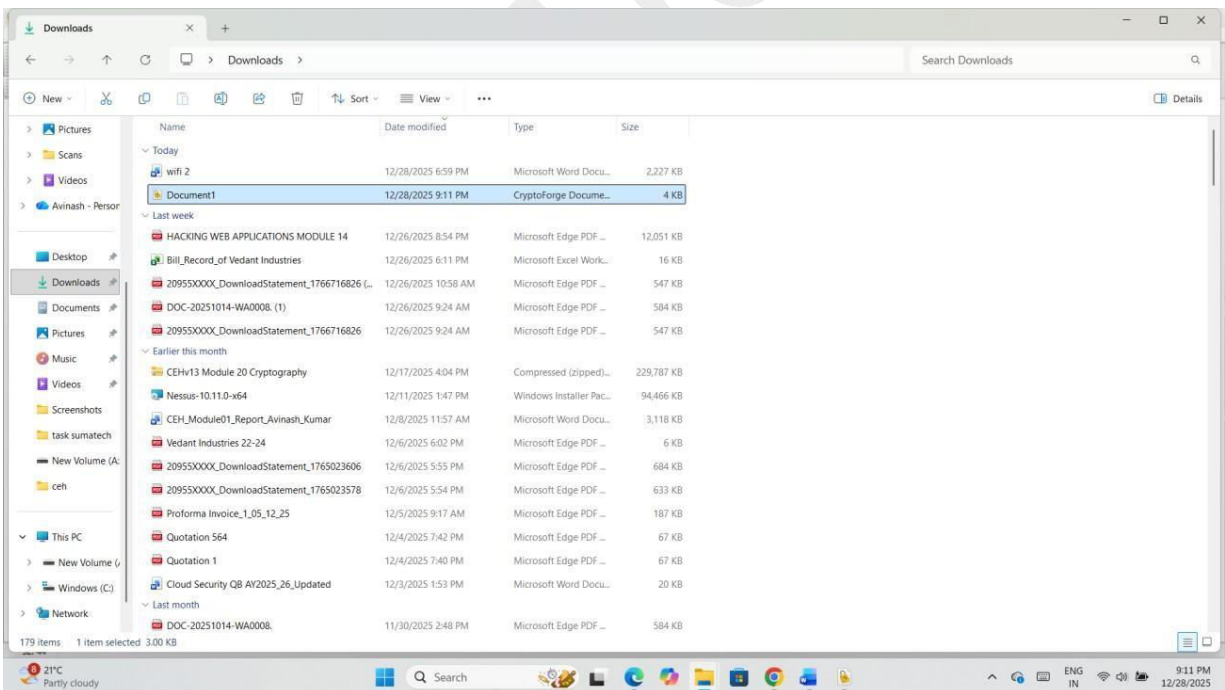
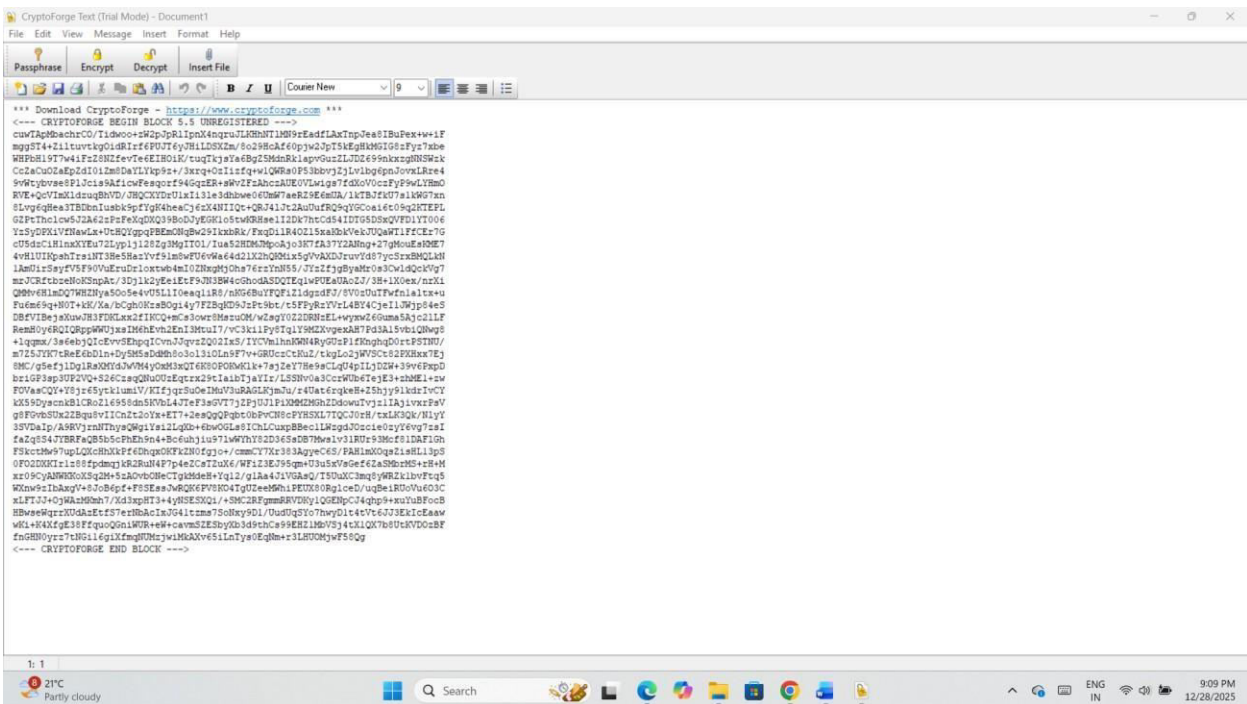
CryptoForge is a file encryption software designed to secure personal and professional data. It enables you to protect sensitive files, folders, and email messages by encrypting them with strong encryption algorithms. Once encrypted, this information can be safely stored on insecure media or transmitted over insecure networks, such as the Internet, while remaining private. Later, the encrypted data can be decrypted back into its original form when needed.



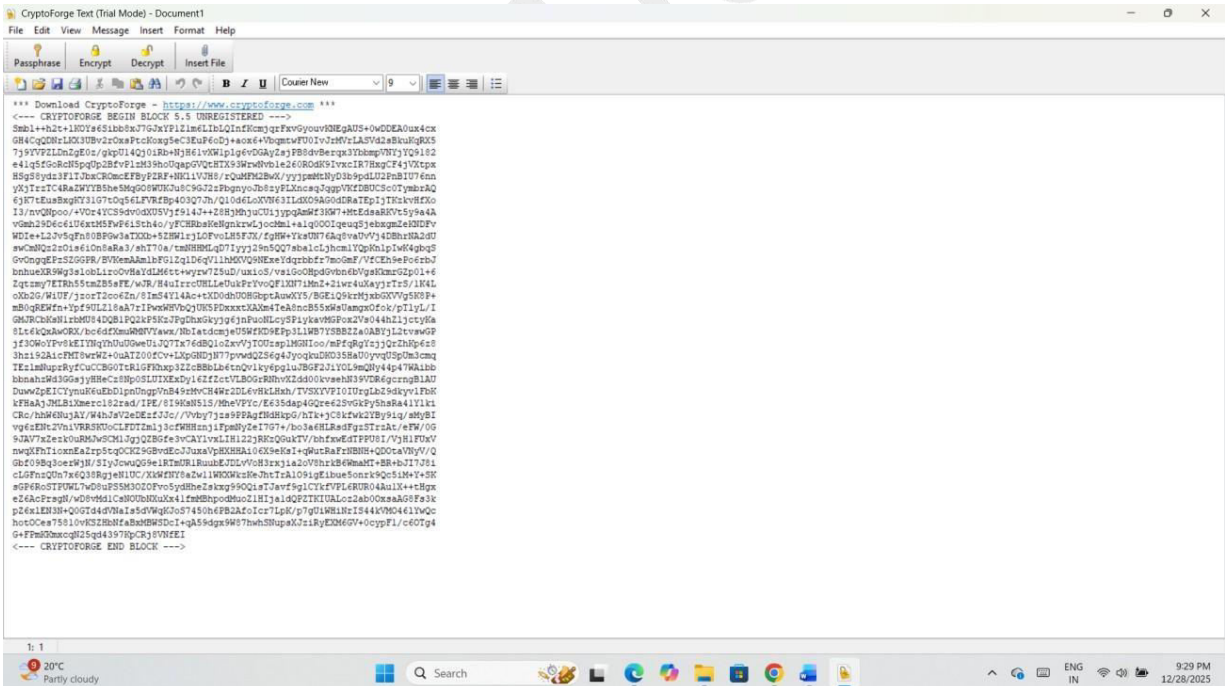
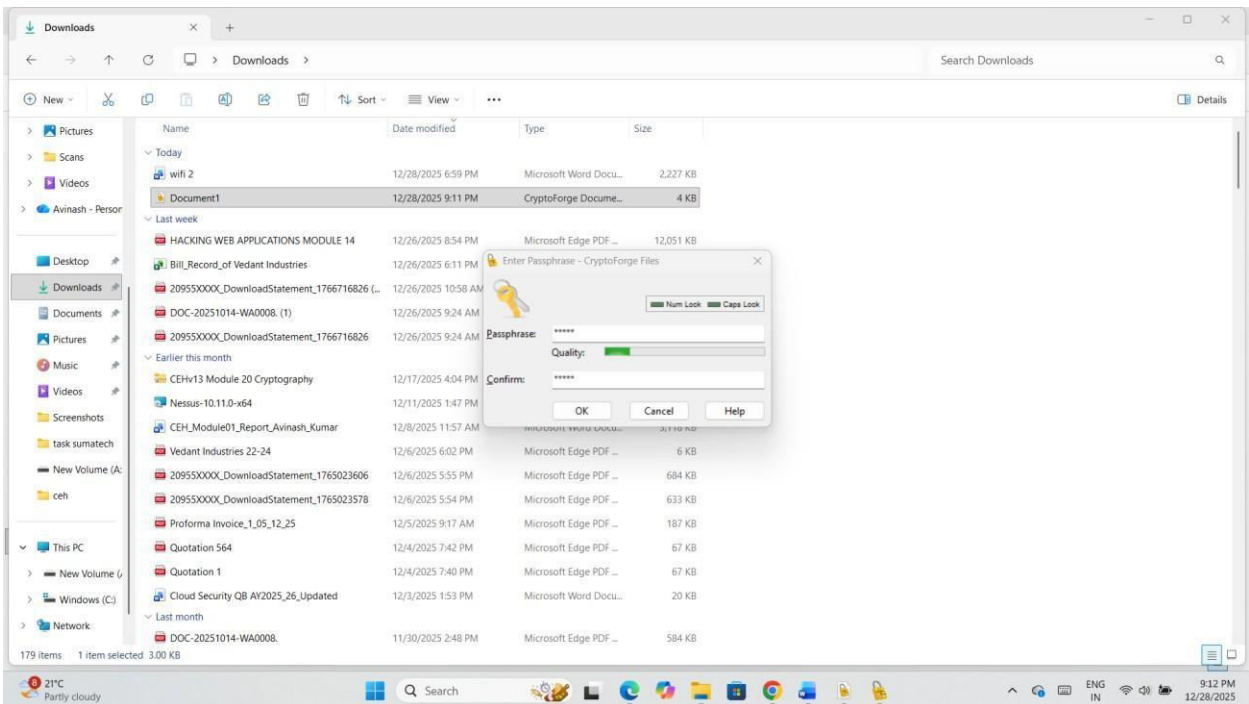
POC'S



POC'S



POC'S



POC'S

